

Summary for ISAC: Yonghui Dong

1. Number and titles of all projects in the lab (including one or two sentences summarizing each project).

There are 4 active projects ongoing on in my lab.

Project 1: Stable isotope labeling assisted construction of tomato spatial metabolomics database.

We are establishing a comprehensive and reliable spatial metabolomics database for tomato. To address the critical challenge of low metabolite annotation rates (typically <10%), we developed a global stable isotope labeling approach. By comparing ^{13}C -labeled and ^{12}C native tomato plants, we can confidently annotate metabolites based on their distinctive isotopic signatures.

Project 2: Host preference and detoxification mechanisms of *Tuta absoluta* on solanaceous plants.

We study how the tomato pinworm, *Tuta absoluta*, chooses its host among solanaceous plants by identifying the attractive volatiles and the insect's detoxification strategies for handling different plant chemicals.

Project 3: Development of a desorption electrospray ionization imaging method for the detection of phytohormones

We are developing a desorption electrospray ionization (DESI) imaging method to visualize plant hormones. This method will use enhanced sample preparation and on-tissue derivatization to overcome the challenges of their low abundance and complex distribution.

Project 4: Development of data analysis software for mass spectrometry imaging

We are creating open-source, vendor-neutral software with a user-friendly interface to streamline the entire mass spectrometry imaging data analysis workflow.

2. Total number of current active grants (including titles and one or two sentences summarizing the objectives of each grant)

0

3. Number of grants received from November 2024-Nov. 2025 (including title)

0

4. Number of grants submitted from Nov. 2024-Nov. 2025(including title)

1. Development of Data Processing Algorithms and Intelligent Analysis Platform for Mass Spectrometry Imaging

5. Number of publications submitted, in press or already published from Nov. 2024-Nov.2025 (including titles of the publication and journals)

I have published 4 papers and submitted 1 manuscript within the period from November 2024 to November 2025.

Published:

1. **Dong, Y.**, Feldberg, L., Wang, Y., Heinig, U., Rogachev, I. and Aharoni, A., 2025. Triple labeling of metabolites for metabolome analysis (TLEMMA): a stable isotope labeling approach for metabolite identification and network reconstruction. The Plant Journal, 123(1), p.e70333.
2. Wang, J., Zhang, L., Wang, S., Wang, X., Li, S., Gong, P., Bai, M., Paul, A., Tvedt, N., Ren, H. **Dong, Y.**, Yang, M., and Guan, Y. 2025. AlphaFold-Guided Bespoke Gene Editing Enhances Field-Grown Soybean Oil Contents. Advanced Science, 12(23), p.2500290.
3. Du, Z., Zhou, Y., Guo, S., **Dong, Y.**, Xu, Y. and Yu, X., 2025. Triterpenoid saponins in tea plants: A spatial and metabolic analysis using UPLC-QTOFMS, molecular networking, and DESI-MSI. Food Chemistry, 475, p.143323.
4. Deis, R., Lerer-Goldshtein, T., Baiko, O., Eyal, Z., Brenman-Begin, D., Goldsmith, M., Kaufmann, S., Heinig, U., **Dong, Y.**, Lushchekina, S. and Varsano, N., 2025. Genetic control over biogenic crystal morphogenesis in zebrafish. Nature chemical biology, 21(3), pp.383-392.

Submitted:

Fu, S., Wang, L., Lu, Y., Heinig, W., and **Dong. Y.**, 2025. Spatial metabolomics reveals tissue-specific metabolic reprogramming during tomato fruit ripening. Plant Physiology.

6. Total number of people working in the lab

6

7. Number of 1 and 2 years of MS.

6

8. Number of 1, 2 and 3 years PhD students

0

9. Number of postdocs and technicians

0

10. A short summary (not more than 500 + 1-2 figs) of one the most exciting project

ShinyCardinal: A Comprehensive Platform for Mass Spectrometry Imaging Data Analysis

While the field of mass spectrometry imaging (MSI) has seen significant growth in recent years, MSI data analysis is still challenging. Advances in improving the mass resolution, spatial resolution, and throughput of MSI has led to a substantial increase in data size. Such large, multidimensional datasets pose significant memory and computational challenges, namely ‘the curse of dimensionality’. An additional obstacle is that the raw MSI datasets are instrument-dependent, sometimes even binary encoded, which hampers MSI data processing and management. Furthermore, MSI datasets are intrinsically complicated. For instance, MSI datasets can suffer from strong pixel-to-pixel spectral variations, and spectra in MSI datasets often contain high-frequency noise and unresolved chemical background.

As such, we introduce ShinyCardinal, an open-source and vendor-neutral software that covers all step in MSI data analysis. It leverages the R package Cardinal to enhance its functionalities by introducing several additional important features, such as the removal of background noises and matrix peaks, deisotoping, absolute quantification, network analysis, and metabolite identification (Fig.1). ShinyCardinal is built as a desktop application (<https://shinycardinal.sourceforge.io>) with a conveniently designed graphic user interface to provide users a stable, consistent, and user-friendly data analysis framework.

To showcase the versatility and functionalities of ShinyCardinal, we analyzed nine MSI datasets obtained from both plant and mammalian samples using different MSI technologies. We manually verified or compared the results with the other MSI software

tools when feasible, showing that ShinyCardinal generated comparable or even superior outcomes.

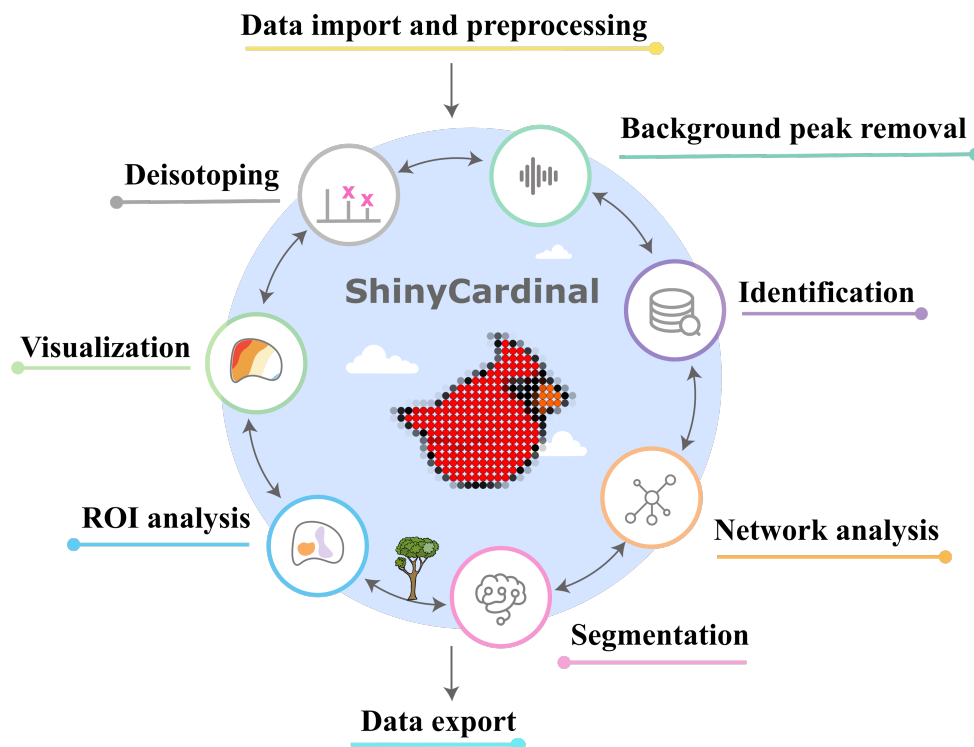


Fig.1 ShinyCardinal modules and the complete data analysis pipeline. ShinyCardinal consists of 11 modules, which cover all the necessary steps for MSI data analysis, i.e., data import, pre-processing, background noise and matrix peak removal, deisotoping, image visualization, regions-of-interest analysis, absolute quantification, image segmentation, network analysis, metabolite identification, and data export.

11. Teaching (including undergraduate student mentoring)

I participated in teaching “Advances in Modern Biology” for undergraduates.

11. Professional and University Services from Nov. 2024-Nov. 2025 (e.g., editorial board service, manuscript and grant reviews, advisory committees, talks/presentation at international and national meetings, various university/center committees or duties)

Manuscript reviewer: I have reviewed 7 manuscripts for the following journals: Bioinformatics (2), Bioinformatics Advances (2), Horticulture Plant Journal (1), Giga Science (1), Rapid Communications in Mass Spectrometry (1)

Conference: I have attended two conferences: (1) **Invited Speaker**, 2nd International Plant Spatio-Temporal Omics Research Conference, Beijing. (2) **Invited Speaker**, Young Scientists Forum on Cutting-Edge Biology, Chongqing University.

University services: I serve as the member of Graduate Employment and Entrepreneurship Committee in our center.

Information needed from Yonghui Dong (Metabolomics platform)

1. Number of labs from the Center that used facility from Nov. 2024-Nov. 2025

69

2. Total budget generated from service

Total revenue: ¥ 748,189

3. Number of outside clients that used facility from Nov. 2024-Nov. 2025

6

4. Number of staff and very briefly their responsibilities

The metabolomics core facility has three staff members.

1. Dr. Xiaxia Wang: oversees GC-MS and LC-MS based metabolomics.
2. Huadong Liu: manages LC-MS targeted analysis.
3. Weijian Zhang: responsible for mass spectrometry imaging.

YONGHUI DONG

EDUCATION

Oct.2010

Jun.2014

● **Trento University, Italy**

Fondazione Edmund Mach, Italy

PhD in Analytical Chemistry

Supervisors: Dr.Pietro Franceschi, Dr.Fulvio Mattivi & Prof.Guella Graziano

Thesis: Mass Spectrometry Imaging: Looking Fruits at Molecular Level

Jun.2012

Dec.2012

● **Max Planck Institute for Chemical Ecology, Germany**

Visiting PhD student

Supervisor: Dr.Ales Svatos

Project: Mass spectrometry imaging of surface lipids on intact *Drosophila melanogaster* flies

Sep.2008

Sep.2010

● **Bologna University, Italy**

Munich University of Technology, Germany

BOKU University, Austria

Triple MSc. in Horticulture Science

Supervisors: Dr. Stefano Tartarini, Dr. Luca Dondini & Prof. Dieter Treutter

Thesis: QTL Analysis of Sugars and Organic Acids in Apricot

Sep.2003

Jun.2006

● **Northwest A&F University, China**

BSc. in Bioengineering

Supervisor: Prof. Zhihui Cheng

Thesis: Effects of UV-C on the control of cucumber powdery mildew

CONTACT

✉ yonghui.dong@gmail.com

 [Github](#)

 [ResearchGate](#)

 [Twitter](#)

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RESEARCH INTERESTS

➤ Mass Spectrometry Imaging

➤ Metabolomics

➤ Stable isotope labeling

➤ Bioinformatics

SKILLS

Programming:

➤ R (9/10)

➤ Python (7/10)

➤ HTML/CSS (7/10)

➤ Git/GitHub (8/10)

➤ Docker (7/10)

Last updated on 2025-12-01.



RESEARCH EXPERIENCE

- June. 2022
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Now

● **Researcher**

Mass Spectrometry Imaging Unit, Weizmann Institute of Science, Israel
- Jan. 2021
|
May. 2022

● **Researcher**

Blavatnik Center for Drug Discovery, Tel Aviv University, Israel
- Dec. 2014
|
Dec. 2020

● **Postdoctoral Research Fellow**

Weizmann Institute of Science, Israel

Supervisor: Prof. Asaph Aharoni
- Jul. 2014
|
Nov. 2014

● **Research Associate**

Trento University, Italy

Supervisor: Prof. Guella Graziano



PUBLICATIONS

- 2025

● **39. Triple labeling of metabolites for metabolome analysis (TLEMMA): a stable isotope labeling approach for metabolite identification and network reconstruction**

The Plant Journal, 123(1), p.e70333

Yonghui Dong, Liron Feldberg, Yi Wang, Uwe Heinig, Ilana Rogachev, and Asaph Aharoni.
- **38. AlphaFold-Guided Bespoke Gene Editing Enhances Field-Grown Soybean Oil Contents**

Advanced Science, 12(23), p.2500290

Wang, Jie, Li Zhang, Shoudong Wang, Xin Wang, Suning Li, Pingping Gong, Mengyan Bai, Yonghui Dong, ..., Yuefeng Guan.
- **37. Triterpenoid saponins in tea plants: A spatial and metabolic analysis using UPLC-QTOFMS, molecular networking, and DESI-MSI**

Food Chemistry 475 (2025): 143323

Du, Zhenghua, Ying Zhou, Shuang Guo, Yonghui Dong, Yongquan Xu, and Xiaomin Yu.
- **36. Genetic control over biogenic crystal morphogenesis in zebrafish**

Nature chemical biology, 21(3), pp.383-392

Deis, R., Lerer-Goldshtein, T., Baiko, O., Eyal, Z., Brenman-Begin, D., Goldsmith, M., Kaufmann, S., Heinig, U., Yonghui Dong, Lushchekina, S. and Varsano, N.,
- 2024

● **35. A cellulose synthase-like protein governs the biosynthesis of Solanum alkaloids**

Science, 386(6728), p.eadq5721

Jozwiak, Adam, Sayantan Panda, Ryota Akiyama, Ayano Yoneda, Naoyuki Umemoto, Kazuki Saito, Shuhei Yasumoto, Yonghui Dong, ..., Asaph Aharoni

- **34. Metabolite Annotation through Stable Isotope Labeling**
TrAC Trends in Analytical Chemistry, 181, p.118037
[Yonghui Dong](#), Liron Feldberg, Asaph Aharoni, and Uwe Heinig
- **33. A Comprehensive Spatially Resolved Metabolomics Dataset for Lampreys**
Scientific Data 11, no. 1: 1075
Gou, Meng, Xiaxia Wang, Xuyuan Duan, Yaocen Wang, Yue Pang, [Yonghui Dong](#) (corresponding author)
- **32. The biosynthetic pathway of the hallucinogen mescaline and its heterologous reconstruction**
Molecular Plant, 17(7), pp.1129-1150
Berman, P., de Haro, L. A., Cavaco, A. R., Panda, S., [Yonghui Dong](#), Kuzmich, N., ... & Aharoni, A.
- **31. New wine in an old bottle: ABCB19, known as auxin exporter, also exports brassinosteroids**
The Innovation Life, 2(2), p.100072
[Yonghui Dong](#), Ning Su, and Yuqin Zhang
- **30. MetaboReport: from metabolomics data analysis to comprehensive reporting**
Bioinformatics, 40(6), p.btae373
[Yonghui Dong](#), and Sergey Malitsky.
- 2023 ● **29. Spatial Metabolomics Reveals the Multifaceted Nature of Lamprey Buccal Gland and Its Diverse Mechanisms for Blood-Feeding**
Communications Biology, 6, 881
Meng Gou, Xuyuan Duan, Jun Li, Yaocen Wang, Qingwei Li, Yue Pang, [Yonghui Dong](#) (corresponding author)
- **28. A complete prostaglandin pathway from synthesis to inactivation in the oral gland of the jawless vertebrate lamprey, Lethenteron camtschaticum**
Developmental & Comparative Immunology
Wang, Yaocen, [Yonghui Dong](#) (co-first author), Xuyuan Duan, Yimu Luan, Qingwei Li, Yue Pang, Feng Sun, and Meng Gou
- **27. Parallel evolution of cannabinoid biosynthesis**
Nature Plants, 1-15
Paula Berman, Luis Alejandro de Haro, Adam Jozwiak, Sayantan Panda, Zoe Pinkas, [Yonghui Dong](#), Jelena Cveticanin, Ranjit Barbole, Rotem Livne, Tali Scherf, Eyal Shimoni, Smadar Levin-Zaidman, Nili Dezurella, Ekaterina Petrovich-Kopitman, Sagit Meir, Ilana Rogachev, Prashant D Sonawane, Asaph Aharoni
- **26. PICA: Pixel Intensity Correlation Analysis for Deconvolution and Metabolite Identification in Mass Spectrometry Imaging**
Analytical Chemistry, 1652–1662
[Yonghui Dong](#), Nir Shachaf, Liron Feldberg, Ilana Rogachev, Uwe Heinig, Asaph Aharoni
- 2022 ● **25. Image to insight: exploring natural products through mass spectrometry imaging**
Natural product reports, 7, 1510-1530
[Yonghui Dong](#), Asaph Aharoni

- **24. TMT-based quantitative proteomics reveals protein biomarkers from cultured Pacific abalone (*Haliotis discus hannai*) in different regions**
Food Chemistry: X, 100355
YimuLuan, [Yonghui Dong](#) (co-first author), Xuyuan Duan, Xiuli Wang, Yue Pang, Qingwei Li, MengGou
- **23. The metabolic and proteomic repertoires of periderm tissue in skin of the reticulated Sikkim cucumber fruit**
Horticulture Research, doi: 10.1093/hr/uhac092
Gulab Chand Arya, [Yonghui Dong](#) (co-first author), Uwe Heinig, Nir Shahaf, Yana Kazachkova, Elinor Aviv-Sharon, Gal Nomborg, Ofir Marinov, Ekaterina Manasheva, Asaph Aharoni, Hagai Cohen
- **22. Quantitative Trait Loci Mapping and Identification of Candidate Genes Linked to Fruit Acidity in Apricot Q13 (*Prunus armeniaca* L.)**
Frontiers in Plant Science, doi: 10.3389/fpls.2022.838370
Luca Dondini, Cecilia Domenichini, [Yonghui Dong](#), Fabio Gennari, Daniele Bassi, Stefano Foschi, Martina Lama, Marco Adami, Paolo De Franceschi, Claudia Cervellati, Lorenzo Bergonzoni, Sara Alessandri, Stefano Tartarini
- **21. Saliva metabolome alterations after acute stress**
Scientific Reports, 12, 18470
Liat Morgan, Rune Isak Dupont Birkler, Shira Shaham-Niv, [Yonghui Dong](#), Tal Wachsman, Lior Carmi, Boris Yakobson, Lihi Adler-Abramovich, Hagit Cohen, Joseph Zohar, Melissa Bateson, Ehud Gazit
- **20. RawHumus: an R Shiny App for Automated Raw Data Quality Control in Metabolomics**
Bioinformatics, btac040
[Yonghui Dong](#), Yana Kazachkova, Meng Gou, Liat Morgan, Tal Wachsman, Ehud Gazit, Rune Isak Dupont Birkler
- 2021 ● **19. CCWeights: an R package and web application for automated evaluation and selection of weighting factors for accurate quantification using linear calibration curve**
Bioinformatics Advances, 1(1), vbab029
[Yonghui Dong](#), Tal Wachsman, Liat Morgan, Ehud Gazit, Rune Isak Dupont Birkler
- **18. Characterization of the PRODUCTION OF ANTHOCYANIN PIGMENT 1 Arabidopsis Dominant Mutant using DLEMMA Dual Isotope Labeling Approach**
Phytochemistry, 186
[Yonghui Dong](#), Liron Feldberg, Ilana Rogachev, Asaph Aharoni
- **17. The GORKY glycoalkaloid transporter is indispensable for preventing tomato bitterness**
Nature Plant, 7, 468-480
Yana Kazachkova, Itay Zemach, Sayantan Panda, Samuel Bocobza, Andrii Vainer, Ilana Rogachev, [Yonghui Dong](#), Shifra Ben-Dor, Dorottya Veres, Christa Kanstrup, Sophie Konstanze Lambert, Christoph Crocoll, Yangjie Hu, Eilon Shani, Simon Michaeli, Hussam Hassan Nour-Eldin, Dani Zamir, Asaph Aharoni
- 2020 ● **16. High mass resolution, spatial metabolite mapping enhances the current plant gene and pathway discovery toolbox**
New Phytologist, 2020, 228:1986-2002.
[Yonghui Dong](#), Prashant Sonawane, Hagai Cohen, Guy Polturak, Liron Feldberg, Shelly Hen Avivi, Ilana Rogachev, Asaph Aharoni

- **15. Rhizosphere microbiome mediates systemic root metabolite exudation by root-to-root signaling**
 Proceedings of the National Academy of Sciences, 2020, 7:3874-3883.
 Elisa Korenblum, [Yonghui Dong](#), Jędrzej Szymanski, Sayantan Panda, Adam Jozwiak, Hassan Massalha, Sagit Meir, Ilana Rogachev, Asaph Aharoni
- 2019 ● **14. Miso: an R package for multiple isotope labeling assisted metabolomics data analysis**
 Bioinformatics, 2019, 35:3524-3526.
[Yonghui Dong](#), Liron Feldberg, Asaph Aharoni
- **13. A Multilevel Study of Melon Fruit Reticulation Provides Insight into Skin Ligno-suberization Hallmarks**
 Plant Physiology, 2019, 179:1486-1501.
 Hagai Cohen, [Yonghui Dong](#), Jędrzej Szymanski, Justin Lashbrooke, Sagit Meir, Efrat Almekias-Siegl, Viktoria Valeska Zeisler-Diehl, Lukas Schreiber, Asaph Aharoni
- **12. In plaque-mass spectrometry imaging of a bloom-forming alga during viral infection reveals a metabolic shift towards odd-chain fatty acid lipids**
 Nature Microbiology, 2019, 3:527-538.
 Guy Schleyer, Nir Shahaf, Carmit Ziv, [Yonghui Dong](#), Roy A Meoded, Eric JN Helfrich, Daniella Schatz, Shilo Rosenwasser, Ilana Rogachev, Asaph Aharoni, Jörn Piel, Assaf Vardi
- 2018 ● **11. DLEMMA-MS-imaging for identification of spatially localized metabolites and metabolic network map reconstruction**
 Analytical Chemistry, 2018, 17:10231-10238.
 Liron Feldberg, [Yonghui Dong](#) (co-first author), Uwe Heinig, Ilana Rogachev, Asaph Aharoni
- **10. Mapping of cell wall aromatic moieties and their effect on hygroscopic movement in the awns of stork's bill**
 Cellulose, 2018, 25: 3827-3841.
 Yael Abraham, [Yonghui Dong](#), Asaph Aharoni, Rivka Elbaum
- 2017 ● **09. TLC surface integrity affects the detection of alkali adduct ions in TLC-MALDI analysis**
 Analytical and Bioanalytical Chemistry, 2017, 409: 5661-5666.
[Yonghui Dong](#), Ruggero Ferrazza, Andrea Anesi, Graziano Guella, Pietro Franceschi
- **08. Engineered gray mold resistance, antioxidant capacity, and pigmentation in betalain-producing crops and ornamentals**
 Proceedings of the National Academy of Sciences, 2017, 14: 9062-9067.
 Guy Polturak, Noam Grossman, David Vela-Corcia, [Yonghui Dong](#), Adi Nudel, Margarita Pliner, Maggie Levy, Ilana Rogachev, Asaph Aharoni
- **07. Impact of tissue surface properties on the desorption electrospray ionization imaging of organic acids in grapevine stem**
 Rapid Communications in Mass Spectrometry, 2016, 30: 711-718.
[Yonghui Dong](#), Graziano Guella, Pietro Franceschi
- 2016 ● **06. More than pictures: when MS imaging meets histology**
 Trends in Plant Science, 2016, 21: 686-698.
[Yonghui Dong](#), Bin Li, Asaph Aharoni

- **05. Sample preparation for mass spectrometry imaging of plant tissues: a review**
Frontiers in plant science, 2016.
[Yonghui Dong](#), Bin Li, Sergey Malitsky, Ilana Rogachev, Asaph Aharoni, Filip Kaftan, Aleš Svatoš, Pietro Franceschi
- **04. Identification of microRNAs and their targets associated with fruit-bagging and subsequent sunlight re-exposure in the “Granny Smith” apple exocarp using high-throughput sequencing**
Frontiers in plant science, 2016.
Dong Qu, Fei Yan, Rui Meng, Xiaobing Jiang, Huijuan Yang, Ziyi Gao, [Yonghui Dong](#), Yazhou Yang, Zhengyang Zhao
- **03. Analytical capabilities of mass spectrometry imaging and its potential applications in food science**
Trends in Food Science & Technology, 2016, 47: 50-63.
Bin Li, Sage JB Dunham, [Yonghui Dong](#), Sohee Yoon, Maomao Zeng, Jonathan V Sweedler
- 2015 ● **02. High production of small organic dicarboxylate dianions by DESI and ESI**
Journal of The American Society for Mass Spectrometry, 2015, 26: 386-389.
[Yonghui Dong](#), Graziano Guella, Fulvio Mattivi, Pietro Franceschi
- 2012 ● **01. Combining intensity correlation analysis and MALDI imaging to study the distribution of flavonols and dihydrochalcones in Golden Delicious apples**
Journal of experimental botany, 2012, 63: 1123-1133.
Pietro Franceschi, [Yonghui Dong](#), Kerstin Strupat, Urska Vrhovsek, Fulvio Mattivi

SOFTWARE

- **01. Miso**
An R package for automated and efficient data analysis workflow to detect the complete repertoire of labeled molecules from multiple-precursor-based labeling experiments.
[GitHub](#), [Scholar](#)
- **02. CCWeights**
An R package and web app for assessing and selecting the best weighting factors (WF) for linear calibration curve-based metabolite quantification.
[GitHub](#), [Scholar](#)
- **03. RawHummus**
An R package and shiny web app for raw data quality control (QC) in metabolomics studies. It generates a comprehensive QC report that enables quick and effortless evaluation of instrument performance and metabolomics data quality.
[GitHub](#), [Scholar](#)
- **04. MetaboReport**
An R package and shiny web app for metabolomics data analysis.
- **05. ShinyCardinal**
An R package and shiny web app for mass spectrometry imaging data analysis.



POSTERS & TALKS

- 2025 ● **ShinyCardinal: an Comprehensive Platform for Mass Spectrometry Imaging Data Analysis (Invited Speaker)**
The Second International Plant SpatioTemporal Omics Research Conference (STOC Plant 2025) (Beijing, China)
- **The use of Stable Isotope Labeling in Plant Metabolomics (Invited Speaker)**
Frontiers in Biology Advances Forum (Chongqing, China)
- 2022 ● **How vampires suck blood? (Talk)**
First Israel Metabolomics Meeting (Israel)
- **Software solutions in untargeted and targeted metabolomics (Poster)**
First Israel Metabolomics Meeting (Israel)
- 2021 ● **Pixel-wise Colocalization Analysis for Metabolite Identification in Mass Spectrometry Imaging (Talk)**
Waterman Seminar Series on Bioinformatics (Germany)
- 2017 ● **Application of MS Imaging in Plant Sciences (Talk)**
Mass Spectrometry Imaging Symposium (France)
- **More than Pictures: When MS Imaging Meets Histology (Talk)**
Spring School for Advanced Imaging in Biological Research (Israel)
- 2015 ● **Mass Spectrometry Imaging of Plant Metabolites (Poster)**
Isranalytica 2015 (Israel)
- 2014 ● **Tissue Surface Properties Jeopardize Quantitative DESI Imaging of Organic Acids in Grapevine Stem (Poster)**
20th International Mass Spectrometry Conference (Switzerland)
- 2013 ● **MS Imaging of Metabolites in Fruits (Poster)**
3rd MS Food day (Italy)
- 2012 ● **MS Imaging of Small Metabolites in Fruits (Talk)**
Ourense Conference on Imaging Mass Spectrometry (Spain)